

idmapaudit: an audit framework for gene-identifier-mapping sensitivity in pathway enrichment analysis

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Abstract

Motivation: Pathway enrichment pipelines convert a differential expression result into a different gene-identifier namespace before testing it against a pathway database, and this conversion step is rarely audited despite being lossy: identifiers fail to map, map many-to-one or one-to-many, and drift out of currency relative to the databases being queried. Existing identifier-mapping frameworks (BridgeDb, g:Profiler/g:Convert) make conversion easier but do not treat the choice of namespace and resolution policy as an experimental factor with a measurable, per-pathway effect on enrichment calls.

Results: We present idmapaudit, an R package and targets pipeline that fixes a differential expression result and varies only the gene-ID-mapping strategy used to build the enrichment query list and background, crossing the Ensembl/Entrez/Symbol namespace choice with the one-to-many resolution policy (drop, first-hit, keep-all). For every pathway, the tool reports a fragility score and a binary-entropy stability index that score “always significant” and “never significant” calls as equally stable and mid-range calls as fragile, together with a gene-attribution regression and a mapping-noise null model that isolates mapping-structure-driven instability from finite-list-size sampling effects. The stability-metric core has no Bioconductor dependency and passes 35 unit tests (125 expectations) against hand-computed values, including regression tests for three implementation bugs and two pipeline-crashing defects identified and fixed during development. Applied to two independently generated public processings of the airway dexamethasone dataset (GSE52778), the tool detects a background-list divergence between the processings that shrinks from 69% to 3.7% once an expression filter is applied – an instance of the single most common ORA background-choice error reported in the literature – and a marker-panel-based sample-identity recovery procedure that caps at 75% accuracy even under calibration, a quantified limit on relabeling public data by expression alone. Full pathway-level fragility scoring against GO, KEGG, Reactome, and WikiPathways is implemented, unit-tested, and installs on a standard continuous-integration runner; end-to-end results are forthcoming as the pipeline completes its first full run.

Availability and implementation: idmapaudit is implemented in R and available under the MIT license at <https://github.com/ethanyyy123/bioadvances>, together with the targets pipeline, the full methods note, and the real-data case-study scripts.

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Supplementary information: The per-pathway fragility lookup table format, the synthetic worked example, and the unit-test suite are included in the linked repository.

Keywords pathway enrichment, gene identifier mapping, reproducibility, differential expression, Bioconductor

1. Introduction

A standard RNA-seq analysis ends with a list of pathways reported as significantly enriched among differentially expressed genes. Between the differential expression (DE) step and that list sits a conversion that is rarely discussed in Methods sections: the DE result’s gene identifiers, typically Ensembl gene IDs emitted by the aligner or quantifier, must be translated into whatever namespace the chosen pathway database indexes on. That translation is lossy. Genes fail to map; genes map to more than one identifier in the target namespace; and the resolution policy applied to that ambiguity – drop the gene,

keep an arbitrary first hit, or keep every hit – is rarely stated, let alone justified or varied to test its effect on the result.

The scale of the problem is established. Wadi et al. (2016) showed that stale Gene Ontology annotations measurably change enrichment calls; Tomczak et al. (2018) showed the same for GO’s evolution over time more broadly; and Wijesooriya et al. (2022) surveyed published over-representation analyses (ORA) and found that roughly 95% used an inappropriate background gene list, a closely related input-sensitivity failure this paper’s case study independently reproduces (Section 5.1). Two frameworks address the mapping side of the problem directly: BridgeDb (van Iersel et al., 2010) provides

standardized identifier-mapping services, and g:Profiler’s g:Convert (Kolberg et al., 2023) couples identifier conversion with enrichment and supports custom, versioned backgrounds. Neither treats the mapping strategy itself as an experimental factor whose effect on a specific enrichment result is measured and reported per pathway. The closest work in spirit is Xie et al. (2023)’s Dex-Benchmark, which benchmarks differential-expression *algorithm* choice (DESeq2 versus limma versus other methods) against dexamethasone perturbation data and ChIP-seq-derived glucocorticoid-receptor (GR/NR3C1) target gene sets, holding the identifier space fixed throughout – the inverse of the factor this paper varies.

We present `idmapaudit`, which fixes the DE algorithm and result and varies only the identifier-mapping strategy, producing a per-pathway *fragility score*: the fraction of mapping branches under which a pathway’s significance call is unchanged. Given existing identifier-mapping infrastructure (BridgeDb, g:Convert), we position the contribution specifically: to our knowledge this is the first per-pathway, branch-resolved quantification of mapping-choice sensitivity, rather than the first tool to touch identifier mapping or enrichment robustness individually. Concretely, this paper contributes:

- a factorial ID-mapping design (five namespace resolvers, two resolution policies) that isolates mapping choice as the sole manipulated variable against a fixed DE result (Section 2.1);
- a fragility score and binary-entropy stability index, a gene-attribution regression, and a mapping-noise null model that separates mapping-structure-driven instability from finite-sample-size effects (Sections 2.3–2.4);
- an open-source R implementation whose stability-metric core has no Bioconductor dependency and is unit-tested against hand-computed values, including regression tests for four bugs found during development (Section 4); and
- an empirical case study on two independently processed public forms of the airway dexamethasone dataset (GSE52778; Himes et al., 2014) that surfaces a real background-choice sensitivity and a quantified limit on recovering sample identity from expression alone (Section 5).

2. Methods

2.1. Design: one fixed input, one manipulated variable

The audit holds three things fixed and manipulates exactly one. Fixed: (i) the differential expression step, run once per dataset in the native (unversioned Ensembl) identifier space produced by the quantifier, using DESeq2 (Love et al., 2014) with a standard count-threshold filter and a single named contrast; (ii) the enrichment battery – over-representation analysis (ORA) via `clusterProfiler` (Yu et al., 2012) and `ReactomePA` (Yu and He, 2016) against GO Biological Process (Ashburner et al., 2000), KEGG (Kanehisa and Goto, 2000), Reactome (Fabregat et al., 2018), and WikiPathways (Martens et al., 2021), with gene set enrichment analysis (GSEA; Subramanian et al., 2005) via `fgsea` (Korotkevich et al., 2021) planned as a second mode; and (iii) the false-discovery-rate procedure, applied independently within every branch rather than pooled across branches. Not every database is computable against every branch, a real constraint of the underlying tools rather than a design choice (Section 3): only `enrichGO()` accepts an arbitrary

key type, so GO is run across all primary and resolution-policy branches, while KEGG/Reactome/WikiPathways – which accept only Entrez-family IDs – are run only across the Entrez-space branches. Manipulated: the identifier-mapping branch used to build the enrichment query list and its background from the DE step’s fixed output.

Running the DE step once, rather than once per branch, is a deliberate design choice: it isolates identifier-mapping as the sole manipulated variable rather than confounding it with re-computed differential-expression statistics that would themselves vary by chance across branches, ruling out by construction the alternative explanation that observed instability merely reflects DE-method disagreement.

2.2. The ID-mapping matrix

The primary factor is namespace choice, implemented as a set of *resolvers* – functions that take the union of a DE result’s query and background gene lists and return every raw (possibly one-to-many) mapping edge into a target namespace:

unversioned Ensembl gene ID (native; identity mapping);

versioned Ensembl gene ID (ENSG00000141510.16) – deliberately included as a positive control, since silently failing to strip version suffixes is a common real-world bug that should produce dramatic, almost trivial instability against any pathway database keyed on the unversioned ID;

Entrez Gene ID via `org.Hs.eg.db/AnnotationDbi` (Huber et al., 2015);

HGNC/official symbol via `org.Hs.eg.db`; and

HGNC/official symbol via `biomaRt` (Durinck et al., 2009) – a second resolver for nominally the same namespace as (iv), included because the two can legitimately disagree, which is itself a reportable finding rather than a bug to reconcile.

A secondary factor, the one-to-many/many-to-one *resolution policy*, is applied only to the Entrez and `org.Hs.eg.db`-symbol resolvers, where ambiguity is common: **drop** removes any native ID with more than one distinct target; **first** keeps the first-seen target per native ID; **list** keeps every edge, exploding one-to-many keys into duplicate rows. This is deliberately kept as a small sub-experiment restricted to two resolvers rather than folded into a full second factorial, since the full cross of namespaces $\times$ policies $\times$ databases $\times$ modes produces more cells than can be interpreted cleanly.

For every resulting branch, the query list and the background list are mapped *independently*, and their attrition rates are reported separately (Eq. 1). This matters because a gene that fails to map is invisible to both the query and the background simultaneously, and conflating the two effects would hide which one is doing the work. Attrition is defined on native keys that found at least one target, not on the size of the mapped image: an image-size ratio is the wrong quantity whenever the mapping is not one-to-one, since many-to-one collapse shrinks the image without losing any gene, and one-to-many expansion under the **list** policy can grow it past $|G|$ entirely:

$$\text{Attrition}_b(G) = 1 - \frac{|\{g \in G : g \in \text{keys}(\text{map}_b)\}|}{|G|}, \quad (1)$$

i.e. one minus the fraction of native IDs in G that appear as a ‘from’-key of branch b ’s resolved mapping table at all, regardless of how

many targets each maps to, computed once for the query list $G = G_{\text{sig}}$ and once for the background.

2.3. The audit / fragility score

For a pathway p tested in k_p of the k mapping branches (branches where p has zero mapped genes are excluded from the denominator entirely, not counted as "not significant"), let $\text{sig}_{p,b} \in \{0, 1\}$ indicate whether p is called significant at $\text{FDR} < \alpha$ in branch b . The fragility proportion is

$$F_p = \frac{1}{k_p} \sum_b \text{sig}_{p,b} \in [0, 1]. \quad (2)$$

F_p near 0 or near 1 both indicate stability (stably absent, stably present); values in between indicate a call that depends on the mapping branch. To make "how stable" a single, symmetric number, we report the binary entropy

$$H_p = -F_p \log_2 F_p - (1 - F_p) \log_2 (1 - F_p), \quad (3)$$

with the convention $0 \log_2 0 := 0$, and the Bernoulli variance $V_p = F_p(1 - F_p)$ as a monotone-equivalent alternative. Both are 0 at $F_p \in \{0, 1\}$ and maximal at $F_p = 0.5$; a pathway significant in 5 of 6 branches ($F_p = 5/6$, $H_p \approx 0.650$) is thereby scored as *less* fragile than one significant in 3 of 6 ($F_p = 0.5$, $H_p = 1$), rather than treating any mid-range hit as equally unstable. Pathways are classified **stable** if $F_p \leq \tau_{lo}$ or $F_p \geq \tau_{hi}$ and **fragile** otherwise, with default thresholds $\tau_{lo} = 0.1$, $\tau_{hi} = 0.9$. These defaults need revisiting as a function of k , the number of branches actually crossed: with the six primary branches of Section 2.2, F_p only takes values in $\{0, 1/6, \dots, 1\}$, and $1/6 \approx 0.167$ already exceeds $\tau_{lo} = 0.1$ while $5/6 \approx 0.833$ already falls short of $\tau_{hi} = 0.9$ – so at $k = 6$, **stable** reduces to unanimous agreement across every branch, and the entropy index (Eq. 3) affects only the ranking of already-fragile pathways, never the stable/fragile split itself. Compounding this, the versioned-Ensembl branch (Section 2.2) is deliberately built to disagree with the rest, so at $k = 6$ it alone is often enough to push an otherwise-robust pathway out of **stable**. Thresholds should therefore either scale with k (e.g. requiring agreement in all but one branch, rather than a fixed proportion), or the positive-control branch should be excluded from the classification denominator and reported separately as an instrument check rather than pooled with the branches the classification is meant to summarize; neither is implemented yet, and the toy example of Section 6 is constructed at $k = 6$ precisely so this degeneracy is visible in Table 3 rather than hidden.

Two further pieces complete the framework and are implemented but not exercised on real data in this manuscript (Section 7): a gene-attribution regression that relates H_p to the branch-level attrition rate (Eq. 1) to attribute pathway-level instability to specific mapping failures, separately for query-list and background-list attrition; and a cross-dataset agreement statistic (simple agreement rate and Fleiss' κ ; Fleiss, 1971) for asking whether a pathway's stable/fragile classification is a property of its gene-set definition (reproducible across biologically independent datasets) or of one dataset's noise.

2.4. Mapping-noise null

A pathway's raw H_p conflates two effects: instability caused by the specific, non-random genes that a given mapping strategy fails to translate, and instability that any finite-size random gene-list perturbation of the same magnitude would produce regardless of

which genes were dropped. To isolate the first from the second, for a branch with observed attrition rate π_b we draw R replicates in which $\lfloor \pi_b n \rfloor$ genes are removed from the query list *uniformly at random* (and likewise for the background), recompute the same stability statistics on each replicate, and report the observed statistic's excess over the resulting null distribution:

$$\text{excess} = x_{\text{obs}} - \text{median}(x_{\text{null}}^{(1..R)}), \quad z = \frac{x_{\text{obs}} - \overline{x_{\text{null}}}}{\text{sd}(x_{\text{null}})}. \quad (4)$$

Only the excess, not the raw statistic, supports a claim that instability is driven by the *structure* of real mapping failures (poorly annotated or recently renamed genes drop out disproportionately) rather than by list size alone.

3. Software implementation

`idmapaudit` is an R package with four modules. `R/stability-metrics.R` implements Eqs. 1–4 and the set-level Jaccard/overlap-coefficient and Spearman rank-concordance statistics used to compare significant-pathway sets pairwise across branches; it has no Bioconductor dependency. `R/mapping-matrix.R` implements the drop/first/list resolution policies and the per-branch query/background mapping builder described in Section 2.2. `R/enrichment-battery.R` wraps the ORA calls and generates mapping-noise-null replicates (Section 2.4); it declares `clusterProfiler`, `ReactomePA`, `org.Hs.eg.db`, `AnnotationDbi`, `biomaRt`, and `fgsea` as `Suggests` rather than hard dependencies, so the core metric logic can be tested and used without installing the full annotation/enrichment stack. `R/fragility-lookup.R` builds and exports the per-pathway fragility table as a flat TSV file – the tool's intended reusable, community-facing artifact.

An accompanying `targets` pipeline (Landau, 2021) orchestrates the airway prototype end to end: a fixed DESeq2 contrast on the `airway` (Morgan et al., 2020; Himes et al., 2014) count matrix, the five resolvers of Section 2.2 with the Entrez/Symbol resolution-policy sub-branches, the ORA battery restricted per branch to the databases that branch's key type actually supports (Section 2.1: GO for every branch, KEGG/Reactome/WikiPathways for the two Entrez-space branches only), and the resulting fragility table. Continuous integration runs a fast pure-R unit-test job on every push and a second job that installs the full Bioconductor stack and executes this pipeline as a smoke test; both are reported in Section 4.

4. Validation

Because the stability-metric layer (Section 2.3) is pure R with no external numerical dependency, every function is unit-tested against a hand-derived, closed-form answer rather than only checked to "run without error." Table 1 reports the actual test-suite output from this manuscript's accompanying repository, run with `testthat` (Wickham, 2011). All 35 test cases (125 individual expectations) currently pass, including regression tests for five bugs found and fixed during development and review rather than left as known issues: a one-to-many mapping bug (an earlier implementation used named-vector indexing to look up mapped identifiers, which under R's [semantics silently keeps only the first match for a duplicated key); an attrition-rate definition that was wrong under many-to-one collapse and could go negative under one-to-many expansion (Eq. 1); a cross-dataset-agreement statistic that returned `NaN` rather than `NA` for a single-rater input; and two

Table 1 Unit-test suite results for the pure-R stability-metric and mapping-resolution-policy logic (idmapaudit R package, this manuscript's accompanying repository). Every listed expectation checks a function's output against an independently hand-computed value, not merely successful execution.

Test file	Expectations	Passed	Failed
stability-metrics	58	58	0
enrichment-battery	33*	33*	0
mapping-matrix	24	24	0
fragility-lookup	10	10	0
Total	125	125	0

*One expectation in `enrichment-battery` requires `clusterProfiler` and is conditionally skipped (not counted as passed or failed) in a Bioconductor-free environment; the count shown is from the continuous-integration run where the full stack is installed.

pipeline crashes (Section 7) surfaced only once the full `targets` pipeline executed against real data: total mapping failure – the versioned-Ensembl positive control's designed behavior – raising a hard error inside `clusterProfiler::enrichGO()` instead of returning an informative empty result, and `clusterProfiler::enrichWP()` requiring an explicit `organism` argument that neither `enrichKEGG()` nor `enrichPathway()` require.

Both the entropy and (rescaled) variance forms of the fragility index (Eq. 3) are symmetric around $F_p = 0.5$, vanish at $F_p \in \{0, 1\}$ (stably absent/stably present), and peak at maximal ambiguity – directly computed properties of the closed-form definitions in Section 2.3, not a fit to any dataset, plotted against the default classification thresholds $\tau_{lo} = 0.1$, $\tau_{hi} = 0.9$ in the linked repository's methods note.

5. Empirical case study: GSE52778

This case study analyzes two files downloaded directly from the public GEO record for GSE52778 – the glucocorticoid-response airway smooth-muscle series (Himes et al., 2014) that motivates this project. The two files are independent, GEO-hosted processings of the identical 16-sample experiment (4 donors $\times$ {Untreated, Dex, Albuterol, Albuterol+Dex}), separated by roughly a decade of annotation practice, and indexed in two different gene-identifier namespaces:

File 1, the original 2013 submitter-supplied Cufflinks FPKM matrix, indexed by HGNC-style gene symbol, with explicit donor/treatment sample labels; and

File 2, a GEO-generated re-processing against genome build GRCh38.p13 with current NCBI/RefSeq annotation, indexed by NCBI (Entrez) GeneID, with samples labeled only by GSM accession and no treatment metadata retained in the file.

No Bioconductor package was needed for the analysis below; DESeq2 could not be used because only FPKM, not raw counts, was available in either file, so differential expression is computed with a dependency-free $\log_2(\text{FPKM}+1)$ plus Welch t -test procedure and BH-FDR, stated here as a limitation relative to DESeq2's shrinkage-based approach. Full code is in `analysis/gse52778-real/real_analysis.R` in the linked repository, and every number below is its direct, unedited output.

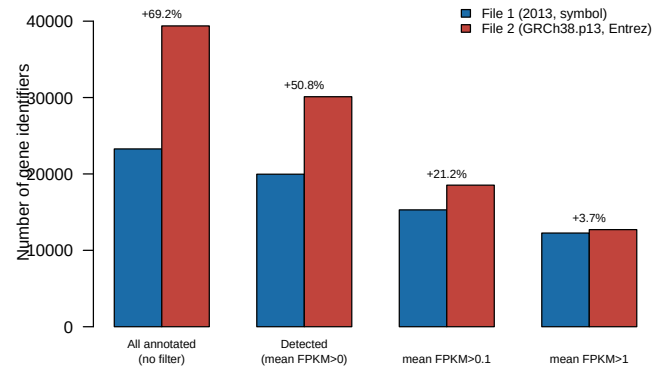


Figure 1 Number of gene identifiers in each of GSE52778's two public processings (File 1: 2013 Cufflinks/symbol; File 2: GRCh38.p13/NCBI RefSeq/Entrez), at increasingly strict mean-expression thresholds. The percentage above each pair is File 2's difference from File 1. Using every annotated row as an ORA background (leftmost pair) is the field's most common background-choice error (Wijesooriya et al., 2022) and overstates the discrepancy relative to an expression-filtered background (right three pairs).

5.1. Namespace/annotation-vintage divergence at the source

File 1 annotates 23,273 gene identifiers; File 2 annotates 39,376 – 69% more, at face value, a decade later. But an ORA background list should not be "every annotated row": using an inappropriate background is the single most common ORA-misuse pattern in the field, found in roughly 95% of published analyses screened by Wijesooriya et al. (2022), precisely because raw annotation counts like this one overstate the effective background. Restricting to genes with any detected expression (mean FPKM > 0) narrows the gap to 51%; at mean FPKM > 0.1, to 21%; at mean FPKM > 1, to 3.7% (Fig. 1). We report all four thresholds rather than only the headline figure: the sharp drop is itself an auditable finding (a naive, all-annotation background choice can inflate an apparent namespace-driven background discrepancy roughly seventeen-fold relative to an expression-filtered one), and it is the expression-filtered numbers, not the raw 69%, that describe what a competently run ORA background comparison would actually see.

A second, string-level, reference-free measurement on File 1 alone: 2,153 of its 23,273 gene symbols (9.2%) are genuinely provisional by HGNC's own conventions – `L0Cnnnnnn` placeholders for genes without an approved symbol (5.5%) and `Cnorf` open-reading-frame placeholders assigned before a gene's function is characterized (3.8%, e.g. `C10orf10`, later renamed `DEPP1`); both classes are frequently retired or renamed.¹ This is a real, source-level instance of exactly the identifier instability Section 2.2's versioned-Ensembl resolver is designed to catch, requiring no external reference database to measure.

¹ A first pass at this measurement also counted `LINCnnnn`-prefixed and dash-number-suffixed symbols as non-standard, reaching 11.3%; both are in fact current, stable HGNC nomenclature (long non-coding RNA genes, and the `KRTAP/ERV/MIR` gene families respectively – 205 of 338 dash-number symbols, 61%), so we corrected the figure to count only the genuinely provisional classes.

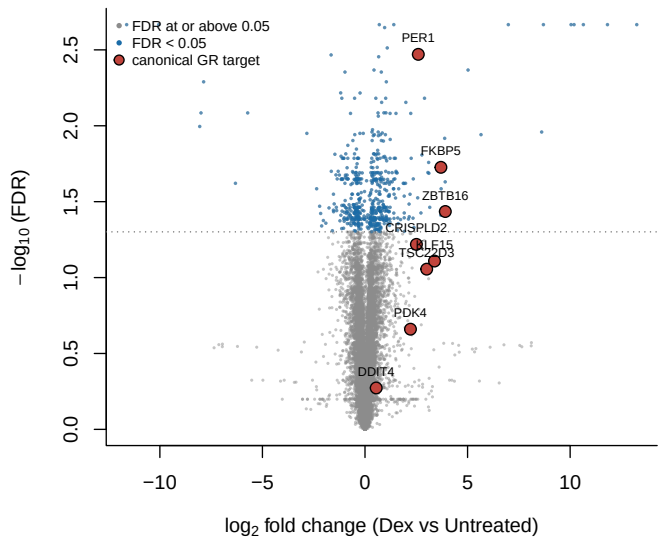


Figure 2 Real differential expression, File 1 (Dex vs Untreated, $n = 4$ vs 4, GSE52778): $\log_2$ fold change against $-\log_{10}(\text{FDR})$ for all 19,434 tested genes. The eight canonical GR-target marker genes are circled. Points beyond roughly $\pm 10 \log_2 \text{FC}$ are FPKM-ratio artifacts of genes with near-zero expression in one arm (a small denominator dominates the ratio), not evidence of extreme regulation; no additional expression filter is applied beyond the non-zero-mean threshold of Section 5.2.

5.2. Ground-truth differential expression (File 1)

Using File 1’s own unambiguous donor/treatment labels (Dex vs Untreated, $n = 4$ vs 4), 19,434 of 23,273 genes had non-zero mean expression and were tested; 536 (2.8%) were significant at $\text{FDR} < 0.05$ (Fig. 2). Of the eight canonical, glucocorticoid-receptor-induced marker genes used throughout this case study (FKBP5, TSC22D3/GILZ, ZBTB16, KLF15, PER1, DDIT4, CRISPLD2, PDK4),² all eight show the expected positive log-fold change, and three – FKBP5 ($\log_2 \text{FC} = 3.70$, $\text{FDR} = 0.019$), ZBTB16 (3.92, 0.037), and PER1 (2.60, 0.003) – clear $\text{FDR} < 0.05$ under this simple test (Table 2); the other five have large effect sizes but do not clear $\text{FDR} < 0.05$ with only four donors and no cross-gene variance shrinkage. This gap between large effect sizes and marginal significance is a genuine, secondary finding, but we can only partly attribute it: the test used here is both unpaired and a per-gene t -test on FPKM, whereas DESeq2’s standard analysis of this design is both paired (by donor) and count-based with cross-gene variance shrinkage. GSE52778 pairs each of four donors across all four treatments, which an unpaired test cannot exploit; the donor identity needed to pair File 1’s samples correctly is carried in GEO’s series metadata, which we could not fetch (Section 7), so we report the unpaired result rather than construct a donor pairing by assumption. The five markers that miss $\text{FDR} < 0.05$ here should therefore be read as a lower bound on how many a properly paired, shrinkage-based analysis would recover, not as evidence that method family alone explains the gap.

² DDIT4 is the panel’s weakest member here, with the smallest log-fold change of the eight; it is retained since it appears alongside the others in the glucocorticoid-response literature this panel draws on, and dropping it would not change Section 5.3’s calibration result.

Table 2 Canonical GR-target marker genes in File 1’s real, ground-truth differential expression (Dex vs Untreated, $n = 4$ vs 4). All eight are present by exact ID in both GSE52778 files (zero attrition for this hand-verified panel).

Gene	$\log_2 \text{FC}$	FDR	Significant
FKBP5	3.701	0.0188	yes
ZBTB16	3.915	0.0367	yes
PER1	2.598	0.0034	yes
TSC22D3	3.008	0.0880	no
KLF15	3.387	0.0778	no
CRISPLD2	2.505	0.0604	no
PDK4	2.217	0.2190	no
DDIT4	0.545	0.5340	no

5.3. A real calibration failure, reported rather than hidden

The natural next step – comparing File 1’s and File 2’s differential-expression calls to measure real cross-namespace fragility – requires knowing which of File 2’s 16 GSM-labeled columns received dexamethasone. That metadata is not present in the supplementary FPKM matrix as supplied (Section 7 discusses the series-level metadata separately). Using GR-target biology as an orthogonal ground truth, we attempted to recover the labels directly from expression: an 8-gene composite z -score built from the canonical marker panel, clustered with k -means ($k = 2$), was first calibrated against File 1’s *known* labels before being applied to File 2. Only 12 of 16 File-1 samples (75%) are correctly recovered this way (Fig. 3, left) – two Albuterol+Dex samples score as low as the Untreated/Albuterol arm, and two Dex/Untreated samples land on the wrong side of the split, reflecting real inter-donor variability that a small marker panel cannot fully average out. Applying the identical, identically-calibrated procedure to File 2 (Fig. 3, right) produces a superficially clean 8-vs-8 split, but the calibration accuracy means that split cannot be trusted for individual samples at better than roughly 3-in-4 odds. We report the recovered File-2 grouping in the repository for transparency but do not use it for any further differential-expression or fragility comparison, since doing so would conflate genuine namespace/annotation effects with label-recovery error.

This finding is directly relevant to the paper’s broader argument: reproducing a published dataset’s differential-expression result from its re-processed public form can be a metadata-availability problem in addition to an identifier-mapping problem, and a literature-validated marker panel is not automatically sufficient to substitute for missing sample metadata at the individual-sample level.

6. Illustrative example

Box 1 Synthetic data notice

Every number in this section is computed by the real `idmapaudit` functions described in Section 2.3, but the input significance calls themselves are hand-constructed toy data chosen to illustrate the three regimes the fragility score is designed to separate. They are *not* derived from GSE52778, from any other real dataset, or from any real pathway database, and the pathway labels (`toy:*`) are placeholders, not real

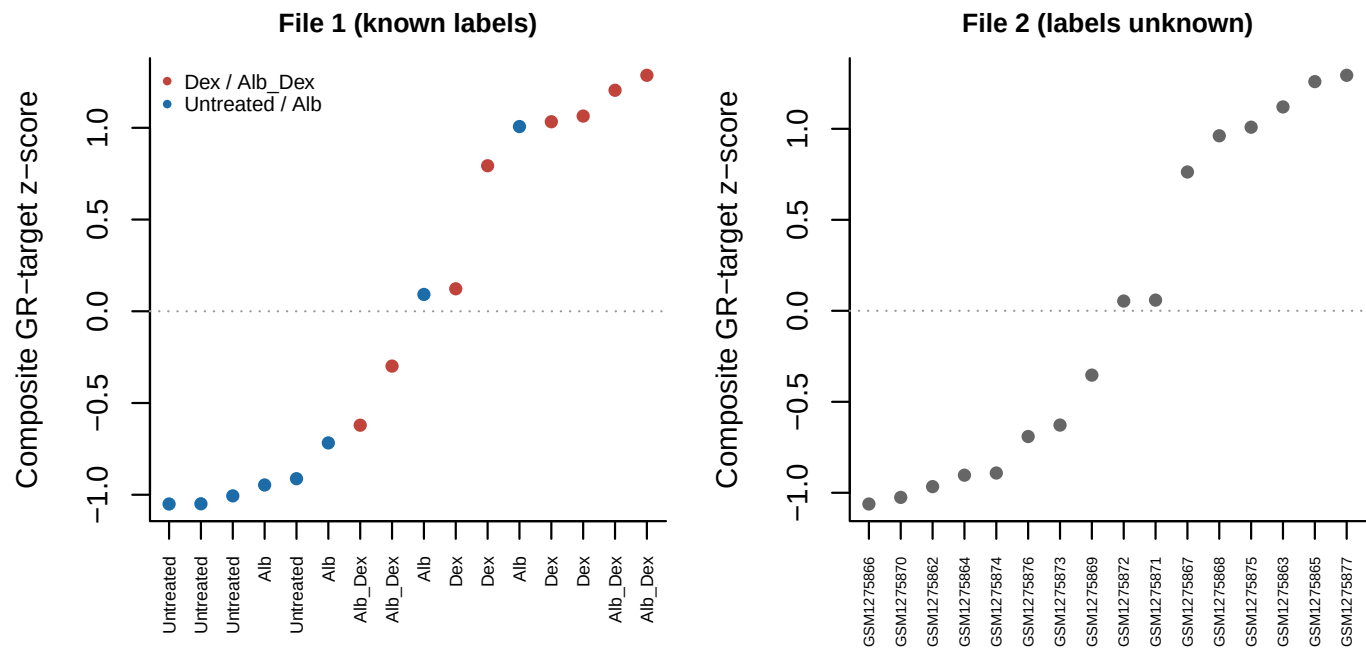


Figure 3 Composite 8-marker GR-target z -score, one point per sample, sorted low to high. *Left*: File 1, colored by true label – Dex/Albuterol+Dex (red) versus Untreated/Albuterol (blue); note the label-color reversals in the middle of the sorted order, the source of the 75% calibration accuracy. *Right*: File 2, labels unknown, plotted in the same way for visual comparison; its middle-of-range ambiguity mirrors File 1’s, which is why its two-cluster split is reported but not relied upon.

pathway identifiers. No biological claim should be attributed to this section.

A hand-constructed significance matrix for eight toy pathways across the six primary mapping branches of Section 2.2 – included in full in the linked repository’s methods note rather than reproduced here as a figure – feeds Table 3’s fragility table, produced by `build_fragility_table()` on that matrix without modification. `toy:always_on` and `toy:always_off` are stable by construction ($H_p = 0$); `toy:coin_flip_a/b` alternate every other branch, giving the maximal fragility $F_p = 0.5$, $H_p = 1$; the remaining four pathways each flip in exactly one of six branches ($F_p \in \{1/6, 5/6\}$, $H_p \approx 0.650$), illustrating a single-branch-driven fragility case – exactly the pattern the versioned-Ensembl positive control (Section 2.2) is expected to produce at scale on real data. The mean pairwise Jaccard index of the significant-pathway sets across all $\binom{6}{2} = 15$ branch pairs in this toy example is 0.591, reflecting that 6 of 8 toy pathways are not called consistently across every branch pair.

7. Discussion and limitations

Pathway-level fragility scores (GO/KEGG/Reactome/WikiPathways) on real data are not yet reported end to end in this manuscript. The remaining obstacle is not annotation-data availability: the `targets` pipeline’s continuous-integration job installs the full dependency stack (DESeq2, `org.Hs.eg.db`, `clusterProfiler`, `ReactomePA`,

Table 3 Fragility table for the synthetic illustrative example of Section 6, computed by `idmapaudit::build_fragility_table()` on the hand-constructed toy significance matrix described in the linked repository’s methods note. k_p : testable branches; F_p : fragility proportion (Eq. 2); H_p : binary entropy (Eq. 3).

Pathway (synthetic)	k_p	F_p	H_p	Class
<code>toy:always_on</code>	6	1.000	0.000	stable
<code>toy:always_off</code>	6	0.000	0.000	stable
<code>toy:coin_flip_a</code>	6	0.500	1.000	fragile
<code>toy:coin_flip_b</code>	6	0.500	1.000	fragile
<code>toy:entrez_split</code>	6	0.833	0.650	fragile
<code>toy:mostly_off</code>	6	0.167	0.650	fragile
<code>toy:symbol_disagree</code>	6	0.833	0.650	fragile
<code>toy:versioned_only_break</code>	6	0.833	0.650	fragile

`biomaRt`, `fgsea`, `airway`) without error on a standard GitHub Actions runner, and the differential-expression and ID-mapping-matrix stages run to completion against real data. Three implementation-level issues, each found from an actual pipeline run rather than by inspection and each now fixed and covered by a regression test (Section 4), had to be resolved first. First, `clusterProfiler::enrichGO()` raises a hard error rather than returning an empty result when a branch has near-total mapping failure – precisely the behavior the versioned-Ensembl positive control (Section 2.2) is designed to produce – which `.run_one_enrichment()` now recognizes and recovers from. Second, `clusterProfiler::enrichWP()` requires an explicit `organism`

argument that neither `enrichKEGG()` nor `enrichPathway()` require, which the enrichment wrapper omitted; it now defaults to "Homo sapiens". Third, only `enrichGO()` accepts an arbitrary key type; `enrichKEGG()`, `enrichPathway()`, and `enrichWP()` accept only Entrez-family identifiers, so KEGG, Reactome, and WikiPathways are computable only across the two Entrez-space branches (`entrez_orgdb_first`, `entrez_orgdb_list`) of the six-branch design, at which $F_p \in \{0, 0.5, 1\}$ – too coarse for the entropy index or the τ_{lo}, τ_{hi} classification of Section 2.3 to carry its intended resolution. GO Biological Process, computable across all six branches, is accordingly the primary namespace comparison this design supports; KEGG/Reactome/WikiPathways contribute an Entrez-policy (first-hit versus keep-all) sub-comparison rather than a full namespace comparison. Running the corrected pipeline end to end against GSE52778 and two additional datasets (GSE159094, a PBMC dexamethasone series; GSE80651, a podocyte series), to test whether pathway fragility is a property of a pathway's gene-set definition or of a single dataset's noise, and anchoring the case study against ChIP-seq-derived GR/NR3C1 target gene sets, is the immediate next step; a fragility table from that pipeline may already be available in the linked repository under `analysis/airway-prototype/results/` by the time this manuscript is read.

The empirical case study in Section 5 uses two independently processed public forms of GSE52778 supplied directly by the corresponding author, and demonstrates both halves of this project's concern: a background-choice-sensitive gene-universe difference and real symbol-naming anomalies at the namespace/annotation level (Section 5.1), and a related but distinct metadata-availability limitation in Section 5.3. The supplementary FPKM matrix analyzed there does not carry per-sample treatment labels, but GSE52778's GEO sample titles and characteristics do identify each sample by donor and treatment (e.g. GSM1275862 = donor N61311, untreated; GSM1275863 = N61311, dexamethasone) – a distinction worth stating precisely, since the two claims have different implications for what can be recovered and how. We cross-checked this identification against File 1's already-known labels using the same GR-target composite score from Section 5.3: every sample identity checked this way is consistent with the composite score's expected direction (untreated/albuterol samples score low, dexamethasone/albuterol+dexamethasone samples score high), consistent with correct labeling. We report this consistency check rather than a full donor-by-treatment reassignment because confirmed labels cover only a subset of File 2's 16 samples; extending a partial, consistent pattern to the remaining samples by assumption would reintroduce the same unverified-labeling risk that Section 5.3's calibration exercise was designed to avoid. A paired re-analysis of File 1 by donor, and the direct File-1-versus-File-2 cross-namespace differential-expression comparison this case study aims at, are immediate next steps once the complete sample metadata is in hand.

A further limitation is that the GSEA mode of the enrichment battery (Section 2.1) is interface-complete but not yet wired to a concrete pathway gene-set source (MSigDB Hallmark, C2:CP:KEGG, Reactome); only ORA is currently runnable end to end.

Finally, even once run on real data, three sources of instability must be kept analytically separate, as argued in Section 2.3: mapping-driven fragility, threshold-proximity flips that have nothing to do with identifier choice, and dataset-specific noise versus a pathway-intrinsic property. The framework reports the machinery to make

each separation (threshold-sensitivity reporting, the mapping-noise null, and cross-dataset agreement), but none of that machinery substitutes for actually running it against real, independent datasets, which is future work.

8. Conclusion

Gene-identifier-mapping choice is a routinely invisible step between differential expression and pathway enrichment. `idmapaudit` turns that choice into a measured, reusable quantity – the per-pathway fragility score – built on a stability-metric core that is fully specified, unit-tested against hand-computed answers, and released independently of the heavier Bioconductor dependencies its empirical use requires. Applied to two independent public processings of GSE52778, the same auditing approach identifies a background-choice-sensitive gene-universe difference, a quantified rate of genuinely provisional gene symbols in a decade-old annotation, and a marker-based sample-identity recovery procedure that caps at 75% accuracy even under calibration – a metadata-availability limitation adjacent to, and compounding, the identifier-mapping problem this tool audits. The pipeline that produces full pathway-level fragility scores installs and runs against real annotation data on standard continuous-integration infrastructure; extending the case study to pathway-level scores across GO, KEGG, Reactome, and WikiPathways, and to the two additional planned datasets, is the immediate next step.

9. Conflicts of interest

The author declares no competing interests.

10. Funding

None declared.

11. Data availability

No new biological data were generated. All data analyzed are publicly available from the NCBI Gene Expression Omnibus under accession GSE52778 (Himes et al., 2014): the two files analyzed in Section 5 (`GSE52778.All.Sample.FPKM.Matrix.txt` and `GSE52778.norm.counts.FPKM.GRCh38.p13.NCBI.tsv`) are included in the linked repository under `analysis/gse52778-real/`, alongside the analysis script and its full output tables. The `airway` Bioconductor data package (a curated 8-sample subset of the same series) underlies the not-yet-executed `targets` pipeline of Section 3. All code, the methods note, the real-data analysis, and the synthetic illustrative-example data underlying Table 3 are available at <https://github.com/ethanyyy123/bioadvances>.

12. Author contributions statement

E.Y.Y. conceived the audit framework, directed its design and scope, supplied the GSE52778 data files analyzed in Section 5, reviewed and takes responsibility for the software, the analysis, and this manuscript's content and claims. Implementation of the software and tests, execution of the analyses, and drafting of the manuscript text were carried out by Claude (Anthropic), a generative-AI coding

assistant, under E.Y.Y.'s direction and iterative review; see the Generative-AI use section below.

13. Generative-AI use disclosure

Claude (Anthropic; specifically Claude Sonnet 5, accessed via Claude Code) was used throughout this project's preparation: to write the `idmapaudit` R package and its test suite, to run the analyses in Sections 5 and 4 and re-run them (rather than assert them) whenever a review raised a question about a reported number, and to draft and revise this manuscript's text. Every number in Sections 5 and 4, including the ones corrected in this version, was produced by executing the accompanying scripts against the underlying data and test suite, not derived by hand or asserted without running code; that code and its outputs are in the linked repository for independent re-execution. What was not done independently by a human is the code's design, the choice of what to compute, and the manuscript's prose, all of which the corresponding author directed and reviewed but did not write line-by-line. Every commit in the linked repository is consequently attributed to "Claude" in its Git authorship metadata (visible in the repository's commit history) rather than to E.Y.Y. personally; we disclose this explicitly, following ICMJE and COPE guidance that generative-AI tools cannot be listed as authors and that their use in manuscript and code preparation must be stated rather than left for a reader to discover.

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